

Supplementary Table S1. Paired repeated-run comparisons of Soft-F2 and weighted BCE

For each metric, values are based on the 10 matched random-seed runs. The paired difference is defined as weighted BCE minus Soft-F2. Bootstrap 95% confidence intervals were obtained from 10,000 resamples of the 10 paired seed-level differences using bootstrap seed 2026. Wilcoxon p-values are two-sided and unadjusted and are presented as supplementary descriptive evidence. For error/loss metrics (Brier score, ECE, and log loss), negative differences indicate lower values under weighted BCE. Probability SD is a dispersion diagnostic rather than a performance score.

Panel A. BeeSpotter

Metric	n	Soft-F2 mean	Weighted BCE mean	Mean difference	Bootstrap 95% CI	W	Two-sided p
F1 (Apis)	10	0.3447	0.3489	0.0042	[-0.0094, 0.0153]	18.0	0.375000
F1 (Bombus)	10	0.0000	0.5221	0.5221	[0.3927, 0.6435]	0.0	0.001953
Macro-F1	10	0.1723	0.4355	0.2632	[0.1979, 0.3227]	0.0	0.001953
Balanced accuracy	10	0.5000	0.5495	0.0495	[0.0316, 0.0663]	1.0	0.003906
ROC-AUC	10	0.5075	0.5947	0.0872	[0.0807, 0.0944]	0.0	0.001953
Average precision	10	0.2108	0.2866	0.0758	[0.0703, 0.0814]	0.0	0.001953
Brier score	10	0.7918	0.2496	-0.5421	[-0.5521, -0.5317]	0.0	0.001953
ECE	10	0.7918	0.2924	-0.4993	[-0.5166, -0.4817]	0.0	0.001953
Log loss	10	12.5938	0.6928	-11.9010	[-11.9261, -11.8777]	0.0	0.001953
Probability SD	10	0.000005	0.081258	0.081253	[0.062429, 0.100337]	0.0	0.001953

Panel B. CIFAR-10 Bird-versus-Frog controlled-imbalance benchmark

Metric	n	Soft-F2 mean	Weighted BCE mean	Mean difference	Bootstrap 95% CI	W	Two-sided p
F1 (Bird)	10	0.3447	0.5637	0.2190	[0.2063, 0.2346]	0.0	0.001953
F1 (Frog)	10	0.0000	0.8639	0.8639	[0.8507, 0.8736]	0.0	0.001953
Macro-F1	10	0.1723	0.7138	0.5415	[0.5305, 0.5522]	0.0	0.001953
Balanced accuracy	10	0.5000	0.7374	0.2374	[0.2266, 0.2497]	0.0	0.001953
ROC-AUC	10	0.4617	0.8199	0.3582	[0.3366, 0.3844]	0.0	0.001953
Average precision	10	0.1961	0.5694	0.3733	[0.3542, 0.3962]	0.0	0.001953
Brier score	10	0.7918	0.1664	-0.6253	[-0.6394, -0.6058]	0.0	0.001953
ECE	10	0.7918	0.1809	-0.6109	[-0.6397, -0.5705]	0.0	0.001953
Log loss	10	12.6127	0.5083	-12.1044	[-12.1390, -12.0594]	0.0	0.001953
Probability SD	10	0.000157	0.258222	0.258066	[0.224714, 0.282477]	0.0	0.001953